

RCF Section 6 — Cosmology
Rewritten Canonical Form — v2.0 (Draft)

Preamble — What This Section Contains and Why

Section 5 examined the saturation regime (black holes) where burden pressure reaches its maximum. Section 6 applies the framework to the universe as a whole — the cosmological regime where Open Expansion operates at cosmic scale and the dark energy mechanism (established in §2.2) produces the observed acceleration.

This section is the v1.0 Section 8 largely intact in architecture, with seven corrections:

1. **The Friedmann equation was asserted without derivation.** The rewrite either provides the explicit computation (burden-metric ADM ansatz restricted to FLRW, variation of S_{eff}) or honestly downgrades to a conjecture with the derivation path stated.
2. **The dark energy mechanism is properly referenced from §2.2.** The v1.0 §8.3.3 introduced dark energy as "SOE frontier pressure" without connecting it to the restorative drive mechanism. The rewrite references the §2.2 derivation ($\rho_{\text{DE}} \sim (H/\Gamma)^2$ with $\Gamma \sim \eta \cdot B$) and applies it cosmologically.
3. **The Γ evolution argument is formalized.** The intuition that " Γ was larger in the past because burden was higher" (established qualitatively in §2.2) is given its quantitative form here: $\rho_{\text{DE}} \sim (1+z)^{-3}$ in matter domination, suppressing early dark energy by $\sim (1+z)^{-6}$ at recombination.
4. **The Theorem 5.4 vs 5.5 misattribution is fixed.** The v1.0 cited "Theorem 5.4" for both the Einstein closure (actually Theorem 4.5.4) and singularity avoidance (actually Theorem 5.2.1). The rewrite uses the corrected anchors.
5. **The Theorem 8.5.1 labeling is corrected.** The v1.0 labeled $\Lambda_B = 0$ as a "Theorem" despite being explicitly conditional on T-2. The rewrite is a Corollary (of Theorem 4.5.4 / Corollary 1.6.4), honestly conditional.
6. **The Q-9 "CLOSED (conditional on T-2)" contradiction is resolved.** The rewrite uses "CONDITIONAL (on T-2)" consistently, reserving "CLOSED" for results with no open blocking dependencies.
7. **The layer-badge collision is resolved.** Section 6 is badged Layer Q' (Quartic — cosmological application of coarse-grained MOE hydrodynamics), consistent with the corrected layer scheme.

§6.0 Purpose: The Universe as a Reconciliation Process

Section 6 applies the framework to the universe as a whole. The central identification: cosmic expansion IS the Open Expansion frontier rate (§1.1), and cosmic acceleration (dark energy) IS the restorative drive of Open Extension (§2.2), evaluated at cosmological scale. The universe is not expanding into a pre-existing vacuum; it is the ongoing process of locality reconciling with causality, with the residual burden (dark energy) being the cost of the global desynchronization of local nows (Theorem 4.2.3).

The section is architecturally disciplined: one-way dependencies on Sections 0–5, explicit Q-item quarantine tags, reaffirmed FIREWALL. The cosmological content is mostly conjectural (the FLRW reduction, the dark sector decomposition, the early-universe dynamics) but is honestly labeled and connected to the framework's established mechanisms.

§6.1 Cosmic Expansion as Open Expansion Rate (Layer Q' — Quartic)

****LAYER Q' - QUARTIC (COSMOLOGICAL APPLICATION)****

The Hubble Rate

> **CONJECTURE 6.1.1 — Expansion as Open Expansion Rate** [Structural conjecture]

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> The Hubble parameter $H(t)$ is the coarse-grained Open Expansion frontier rate:

 $\succ$
$$H(t) = \gamma_{\text{eff}}(t) = (1/V(t)) \cdot dV/dt \quad (6.1.1)$$

>

> where $V(t)$ is the reconciled causal volume and $\gamma_{\text{eff}}(t)$ is the effective Open Expansion rate at cosmological scale.

 $\succ$

> The universe expands because new open expansions continuously incorporate new events at speed $c = \gamma \cdot \ell_0$ (Theorem 1.1.2). $H(t)$ is the macroscopic signature of this microscopic process.

The Expansion Rate Limit

> **COROLLARY 6.1.2 — Expansion Rate Limit** [Established (conditional)]

>

$$H(t) \leq \gamma_{\max} = 1/\varepsilon \quad (6.1.2)$$
 $\succ$

> The Hubble rate is bounded above by the maximum Open Expansion flux capacity. This is a genuine prediction: the expansion rate cannot exceed the microscopic reconciliation rate.

 $\triangleright$

> ***Note:** This is a Corollary (not a Conjecture) — it follows directly from Theorem 1.1.2 (finite propagation rate) applied at cosmological scale. The v1.0 "Corollary 8.1.2" labeling is preserved but the derivation is now cleanly grounded in §1.1.

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requires averaging this non-local channel. The averaging is not addressed in the v1.0; the rewrite flags it as part of the open derivation.

The Forward-Reference Resolution

> **REMARK 6.2.3 — Forward References Resolved** [Structural]

>

> Conjecture 6.2.2 resolves the following forward references:

> - The $\Lambda_B = 0$ forward reference from §4.5.5 (cosmological constant vanishes, no Λ term in Friedmann).

> - The dark energy reinterpretation from §2.2 (dark energy is the residual burden pressure, not a cosmological constant).

>

> The cubic volume $\leftrightarrow$ scale factor connection (v1.0 "§2.10 forward reference") is resolved by Conjecture 6.2.1, referencing Definition 3.6.2 (not the non-existent "Definition 2.10" of v1.0).

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§6.3 The Dark Sector (Layer Q' — Quartic)

Three-Part Decomposition

> **DEFINITION 6.3.1 — Dark Sector Decomposition** [Established]

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- > The dark sector decomposes into three parts:

 $\succ$

> 1. ****Primary Dark Matter (relational burden):**** $\Theta^{(\text{rel})} = [\hat{C}_{\alpha}, \Pi_{\text{net}}]$ (Definition 4.4.1). DERIVED, not quarantined. Resolves the §4.4 forward reference. Q-1 (dark matter amplitudes) remains OPEN — the structural match is established, quantitative recovery requires numerical simulation.

 $\succ$

> 2. ****Cross-Sector Gravity (QUARANTINED).**** The hypothesis that distinct sectors gravitationally interact through their branch weights. QUARANTINED because it crosses the FIREWALL (Principle 2.8.1). Isolated from the deductive stack.

>

> 3. ****Dark Energy (residual burden pressure):**** The dynamic residual of the Open Extension restorative drive (§2.2). NOT a cosmological constant ($\Lambda_B = 0$, Corollary 4.5.5). The mechanism is established in §2.2; the cosmological application is below.

Dark Energy: Cosmological Application

> **CONJECTURE 6.3.2 — Dark Energy as Restorative Drive (Cosmological)**** [Structural; mechanism from §2.2]

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> The effective dark energy density is

 $\succ$
$$p_{DE}(t) \sim (H(t) / \Gamma_{eff}(t))^2 \cdot p_{source} \quad (6.3.2)$$

- >
- > where (from §2.2):
- > - $H(t)$ is the Hubble rate (Conjecture 6.1.1)
- > - $\Gamma_{\text{eff}}(t) \sim \eta \cdot B_{\Delta}(t)$ is the burden-dependent relaxation rate (Property 2.2.2)
- > - ρ_{source} is the saturation burden density (set by Π_{max} and ℓ_0)
- >
- > The quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch ($B_{\Delta} = \|\text{mismatch}\|^2$).

The 120-Order Resolution

- > ****THEOREM 6.3.3 — Cosmological Constant Problem Resolved**** [Established (structural)]
- >
- > The 120-order-of-magnitude discrepancy does not arise in RCF.
- >
- > **Argument:**
- > 1. There is no vacuum energy ($\Lambda_B = 0$, Corollary 4.5.5). The framework does not quantize fields on a fixed background; the ground state has zero burden.
- > 2. Dark energy is not a vacuum energy; it is the rate of an ongoing process (the restorative drive).
- > 3. The natural scale is $\rho_{\text{DE}} / \rho_{\text{Planck}} \sim (H/\Gamma)^2$. For $\Gamma \sim 1/t_P$ (the natural microscopic rate):
- >
- > $\rho_{\text{DE}} / \rho_{\text{Planck}} \sim (H_0 \cdot t_P)^2 \sim (10^{-61})^2 \sim 10^{-122}$
- >
- > This is the observed value. The 120-order suppression is $(H_0/\Gamma)^2 = (\ell_P/R_H)^2$ — the square of the IR/UV scale ratio. The exponent 2 is not tuned; it comes from burden being quadratic in mismatch.
- >
- > **Status:** [Established (structural)]. The 120-orders problem is dissolved as a category error: there is no vacuum energy to fine-tune. The dark energy density is a dynamical residual, not a constant.

The Γ Evolution and Early Dark Energy Suppression

- > ****THEOREM 6.3.4 — Early Dark Energy Suppression**** [Established (derived)]
- >
- > The burden-dependent relaxation rate $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$ was larger in the past (higher burden, steeper gradient). In matter domination:
- >
- > - $H \sim (1+z)^{3/2}$ (standard Friedmann)
- > - $B_{\Delta} \sim \rho_{\text{matter}} \sim (1+z)^3$ (burden tracks energy density, since burden sources gravity, §4.3)
- >
- > Therefore:
- >

> *Note:* This is the cosmological restatement of the Convergence Theorem (Theorem 1.6.1) applied at global scale. The v1.0 "Conjecture 8.3.3" status is upgraded: the asymptotic decay is established (from the Convergence Theorem); only the present-day amplitude requires T-2.

§6.4 $\Lambda_B = 0$ (Restated, Corrected)

> **COROLLARY 6.4.1 — $\Lambda_B = 0$ (Cosmological Restatement)** [Conditional on T-2]**

> The burden-derived cosmological constant is exactly zero:

$$> \Lambda_B = 0 \quad (6.4.1)$$

> This restates Corollary 4.5.5 (and Corollary 1.6.4) cosmologically. It is a structural consequence of Master-Zero being an exact asymptotic attractor, not a fine-tuning.

> *Status:* [CONDITIONAL on T-2]. The v1.0 labeled this "Theorem 8.5.1" despite being explicitly conditional on T-2. The rewrite is a COROLLARY (of Theorem 4.5.4 / Corollary 1.6.4), honestly conditional. A theorem conditional on an unproved conjecture is not a theorem; it is a conditional proposition.

> *The v1.0 "Theorem 5.3.1 / 5.5" reference* is corrected: the underlying result is Theorem 4.5.4 (Einstein closure) and Corollary 4.5.5 ($\wedge_B = 0$), both in §4. The non-existent "Theorem 5.3.1" is removed.

> **REMARK 6.4.2 — Q-9 Status** [Structural]

> Q-9 (cosmological constant problem) status: ****CONDITIONAL (on T-2)****.

$\neg \wedge B = 0$ is structurally derived (Corollary 6.4.1, conditional on T-2).

- > - The 120-order discrepancy is structurally resolved (Theorem 6.3.3).

- > - The dark energy mechanism is established (§2.2, §6.3).

- > - The formal closure ($\wedge B = 0$ as a proven theorem, not a conditional) requires T-2.

> The v1.0 "CLOSED (conditional on T-2)" was self-contradictory — "CLOSED" and "conditional on an open target" cannot both hold. The rewrite uses "CONDITIONAL (on T-2)" consistently, reserving "CLOSED" for results with no open blocking dependencies.

§6.5 The Early Universe (Layer Q' — Quartic)

The Initial Kinematic State

- > ****CONJECTURE 6.5.1 — Initial Kinematic State**** [Structural conjecture]
- >
- > The universe began in a state of maximal burden: ω_{kin} with B_{Δ} near saturation ($B_{\Delta} \approx B_{\text{max}}$). The universe starts maximally unreconciled; Open Extension descent drives it toward $\ker(\hat{M})$.
- >
- > This resolves the §1.6 forward reference (cosmological initial condition): the initial state is the boundary condition, not a derived result. The framework does not derive why the universe started in this state; it assumes it as the cosmological initial condition.

Inflation

- > ****CONJECTURE 6.5.2 — Inflation as Rapid Open Expansion**** [Structural conjecture; Q-3 OPEN]
- >
- > Cosmic inflation was the initial phase of maximal Open Expansion at rate γ_{max} , with Open Extension descent suppressing cross-sub-sector coherence.
- >
- > **Status:* Q-3 (inflation) OPEN. Per Spec Ch. 12.2, inflation is "separate from the core framework" — the inflationary epoch is a boundary condition on the cosmological initial state, not a theorem-target of the relational algebra. The framework provides a structural mechanism (rapid Open Expansion + MOE dephasing) but does not derive the inflationary observables (n_s , r , f_{NL}).

Sector Formation Epochs

- > ****CONJECTURE 6.5.3 — Sector Formation Epochs**** [Structural conjecture; Q-4 PARTIAL]
- >
- > As Open Extension descent crossed decoherence thresholds ($B_{\text{cross}} \gg B_{\text{intra}}$, Theorem 2.6.3), record sectors formed. The matter-antimatter asymmetry (Q-4) is fixed at this epoch.
- >
- > **Status:* Q-4 PARTIAL. T-4 (Born rule) is CLOSED (§2.7). The mechanism (sector formation via decoherence) is proposed. The formal derivation of the asymmetry amplitude δp is open — the framework explains preservation of an existing asymmetry, not its origin.

§6.6 Λ CDM Recovery and Guardrails (Layer Q' — Quartic)

Λ CDM Recovery Status

- > ****REMARK 6.6.1 — Q-13 Status**** [Structural]
- >

> Q-13 (Λ CDM recovery) status: ****PARTIAL****.
 >
 > - FLRW form: established (Conjecture 6.2.2, conditional on the derivation path).
 > - Dark sector mechanisms: established (relational burden = dark matter, restorative drive = dark energy).
 > - Amplitudes (H_0 , Ω_m , Ω_{DM} , $\Omega_{\Lambda,eff}$): OPEN. Require numerical simulation of Open Extension descent on FLRW with realistic K_ω .
 > - The framework predicts deviations from Λ CDM ($w > -1$, evolving; $\rho_{DE} \rightarrow 0$ as $t \rightarrow \infty$), which are testable.

Lorentz Invariance in the Deep IR

> ****REMARK 6.6.2 — Q-17 Status**** [Structural]
 >
 > Q-17 (Lorentz invariance in deep IR) status: ****PARTIAL****.
 >
 > - Speed limit $c = \gamma \cdot \ell_0$: established (Theorem 1.1.2).
 > - Lorentzian signature $(-, +, +, +)$: established (Theorem 4.5.3, sufficiency).
 > - Full Lorentz group equivalence (boosts, velocity addition, relativistic momentum): OPEN, requires T-1 (Lorentz factor derivation).

Nine Cosmological Guardrails

> ****GUARDRAILS 6.6.3**** [Structural]
 >
 > 1. **** $\Lambda_B = 0$ is exact (conditional on T-2).**** No vacuum energy; dark energy is dynamic residual pressure.
 > 2. ****Dark energy is not a cosmological constant.**** It decays as $t \rightarrow \infty$ (Corollary 6.3.6).
 > 3. ****No ad hoc scalar inflaton fields.**** Inflation is Open Expansion at γ_{max} (Conjecture 6.5.2); no new primitive fields.
 > 4. ****Cosmic expansion dynamics derive from Open Expansion / Open Extension competition.**** Not from a Λ term.
 > 5. ****The FIREWALL is reaffirmed.**** Branch weights (Layer A) do not source curvature (Layer Q'). Cross-sector gravity is quarantined.
 > 6. ****Dark matter is relational burden.**** Not a new particle; a structural commutator cost.
 > 7. ****The early universe had higher Γ .** Early dark energy is suppressed (Theorem 6.3.4).
 > 8. ****The Hubble tension may reflect local B_Δ variation.**** Local Open Extension rate may differ from the global average inferred from the CMB.
 > 9. ****All cosmological amplitudes require numerical simulation.**** The structural mechanisms are established; the quantitative predictions are open (Q-1, Q-2, Q-13).

§6.7 Architectural Summary

What Section 6 Contains

| Object | Definition | Status |

|-----|-----|-----|
 | Hubble rate = Open Expansion rate | Conj 6.1.1 | Structural conjecture |
 | Expansion rate limit | Cor 6.1.2 | Established (conditional) |
 | Cubic volume growth | Conj 6.2.1 | Structural conjecture |
 | Friedmann equation | Conj 6.2.2 | Structural conjecture (derivation path stated) |
 | Dark sector decomposition | Def 6.3.1 | Established |
 | Dark energy (cosmological application) | Conj 6.3.2 | Structural (mechanism from §2.2) |
 | 120-order resolution | Thm 6.3.3 | Established (structural) |
 | Early dark energy suppression | Thm 6.3.4 | Established (derived) |
 | Dark energy equation of state | Thm 6.3.5 | Established (derived) |
 | Asymptotic decay of dark energy | Cor 6.3.6 | Established |
 | $\Lambda_B = 0$ (cosmological) | Cor 6.4.1 | Conditional on T-2 |
 | Initial kinematic state | Conj 6.5.1 | Structural conjecture |
 | Inflation | Conj 6.5.2 | Q-3 OPEN |
 | Sector formation epochs | Conj 6.5.3 | Q-4 PARTIAL |

What Section 6 Does NOT Contain (Deferred)

Object	Deferred to	Reason
Formal action S_{eff}	§7	Audit/closure
Numerical simulation of MOE on FLRW	Future work	Closes Q-1, Q-2, Q-13
Lorentz factor derivation	T-1	Spectral analysis
Standard Model recovery	Q-6	Quarantined

The Emergence Ladder Through Section 6

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From §0–§5:

[Full emergence chain through black holes]

§6 (Cosmology — Layer Q', cosmological application):

$H(t)$ = Open Expansion frontier rate

- Friedmann from Open Extension on FLRW (conjectural, derivation path stated)
- Dark sector: relational burden (DM) + restorative drive (DE) + cross-sector (quarantined)
- 120-order resolution ($\rho_{\text{DE}} \sim (H/\Gamma)^2$, no vacuum energy)
- Γ evolution → early dark energy suppressed ($\Omega_{\text{DE}} \sim 10^{-18}$ at recombination)
- Equation of state: $w > -1$, evolving from 0 to -1
- Asymptotic decay: $\rho_{\text{DE}} \rightarrow 0$ as $t \rightarrow \infty$
- $\Lambda_B = 0$ (conditional on T-2)
- Initial state: maximal burden (conjectural)
- Inflation: rapid Open Expansion (Q-3 OPEN)
- Sector formation (Q-4 PARTIAL)

→ Section 7: Audit, formal action S_{eff} , closure

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Acyclicity Test

Question: Does Section 6 define every object before using it? Are there forward references to §7+ objects?

Answer: The chain is acyclic within Section 6.

- $H(t)$ (§6.1) uses Open Expansion (§1.1) and γ (§1.1.2).
- The Friedmann equation (§6.2) uses the burden tensor (§4.3), the ADM ansatz (§4.5.1), $\Lambda_B = 0$ (Corollary 4.5.5). No forward references.
- Dark energy (§6.3) uses the mechanism from §2.2 ($\rho_{DE} \sim (H/\Gamma)^2$), Γ evolution (Property 2.2.2), burden (§1.3).
- $\Lambda_B = 0$ (§6.4) restates Corollary 4.5.5 / Corollary 1.6.4.
- The early universe (§6.5) uses the initial state (conjectural) and sector formation (§2.6).

No forward references to §7+ objects. The formal action S_{eff} is deferred to §7.

All stale cross-references corrected:

- "Theorem 5.4" (for Einstein closure) → "Theorem 4.5.4"
- "Theorem 5.4" (for singularity avoidance) → "Theorem 5.2.1"
- "Theorem 5.3.1 / 5.5" → "Theorem 4.5.4 and Corollary 4.5.5"
- "Definition 2.10" / "§2.10" → "Definition 3.6.2" / "§3.6"
- "§0.3 Property 3" → "§1.3.2 (Property 1.3.2, Burden Linearity)"

Verdict: Section 6 is acyclic. The architecture holds through cosmology.

End of Section 6 — Rewritten Canonical Form v2.0 (Draft).

This draft applies seven corrections to v1.0 Section 8:

The Friedmann equation derivation is honestly stated. Conjecture 6.2.2 provides the derivation path (FLRW reduction of the ADM ansatz, S_{eff} variation) but acknowledges the computation hasn't been performed. The $8\pi G/3$ coefficient is imported from standard GR via the κ_B calibration (Theorem 4.6.1).

The dark energy mechanism is properly referenced from §2.2. The v1.0 introduced dark energy as "SOE frontier pressure" without connecting it to the restorative drive. The rewrite references the §2.2 derivation and applies it cosmologically.

The Γ evolution argument is formalized. Theorem 6.3.4 gives the quantitative form: $\rho_{DE} \sim (1+z)^{-3}$ in matter domination, suppressing early dark energy by $\sim (1+z)^{-6}$ at recombination. The tension with CMB bounds vanishes.

****The Theorem 5.4 vs 5.5 misattribution is fixed.**** All references use the corrected anchors (Theorem 4.5.4 for Einstein closure, Theorem 5.2.1 for singularity avoidance).

****The Theorem 8.5.1 labeling is corrected.**** $\Lambda_B = 0$ is a Corollary (Corollary 6.4.1), not a Theorem. It is honestly conditional on T-2.

****The Q-9 "CLOSED (conditional on T-2)" contradiction is resolved.**** The status is "CONDITIONAL (on T-2)" consistently, reserving "CLOSED" for results with no open blocking dependencies.

****The layer-badge collision is resolved.**** Section 6 is badged Layer Q' (Quartic — cosmological application), consistent with the corrected layer scheme.

The dark energy mechanism now has its full cosmological application: the 120-order resolution (Theorem 6.3.3), the early dark energy suppression (Theorem 6.3.4), the equation of state (Theorem 6.3.5), and the asymptotic decay (Corollary 6.3.6). The quantitative amplitude (present-day ρ_{DE}) still requires T-2, but the structural predictions are testable against DESI/Euclid.